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Data Science

AutiWise - A comprehensive digital toolkit for individuals with Autism Spectrum Disorder

Group No. 10

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Project Guide

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Abstract

AutiWise is an AI-powered digital system designed to provide early detection and continuous support for individuals with Autism Spectrum Disorder (ASD). It integrates four modules: a diagnostic tool for identifying ASD indicators, a conversational chatbot for communication and emotional support, a voice-command routine tracker to reduce anxiety and improve time management, and a social skills coach that generates AI-driven conversational scripts for role-play and learning. By combining AI with an intuitive interface, AutiWise aims to support individuals with ASD through timely intervention, improved communication, and enhanced social inclusion.

Sustainable Development Goals(SDG) mapped

- SDG 3: Good Health and Well-being - Promotes early detection and continuous support for individuals with autism, improving mental health and quality of life.

Introduction

- Autism Spectrum Disorder (ASD) is a condition characterized by challenges in social interaction, communication, emotional regulation, and repetitive behavioral patterns.
- Autism Spectrum Disorder (ASD) is increasing worldwide, impacting communication, social interaction, and daily routines of affected individuals.
- This creates a strong need for an affordable, AI-powered, and interactive system that can deliver timely detection and continuous support for individuals with ASD.

Motivation

- Many individuals with ASD still struggle with expressing emotions, building social connections, and maintaining structured routines in daily life.
- Existing digital solutions often address only one aspect (diagnosis, communication, or routine management), lacking integration.

Objectives

- To design and implement an AI-powered system for the early detection of Autism Spectrum Disorder (ASD).
- To develop a conversational chatbot that facilitates communication support and emotional expression for individuals with ASD.
- To build a voice-command-based routine tracker that helps users create and follow structured daily schedules to reduce anxiety and improve time management.
- To create a social skills coaching module to generate real-world conversation scripts for role-play and social learning.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Gap Identified
1	Using Artificial Intelligence to Improve Empathetic Statements in Autistic Adolescents and Adults: A Randomized Clinical Trial [1]	Koegel, L.K., Ponder, E., Bruzzese, T. et al	2025	The paper uses a methodology of LLM-based AI hat prompts users and gives feedback on empathetic responses	Focuses only on empathetic verbal responses; lacks integration with daily schedules
2	Examining Virtual Reality Interventions for Social Skills in Children with Autism Spectrum Disorder: A Systematic Review [2]	Altın, Y., Boşnak, Ö. & Turhan, C	2025	Systematic review of VR social skills interventions in children with ASD	VR hardwares are expensive. VR environments can cause autistic person more anxiety, disorientation, or cybersickness

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Gap Identified
3	AI-Powered Autism Detection, Awareness, and Social Integration: A Comprehensive Framework [3]	R. S. Ali Fathima, R. Gokkul, M. Nirmalsahaj, S. Rohith and S. Rupes	2025	This paper uses rule-based chatbot to provide conversational support	The chatbot is unable to answer outside its scope
4	mHealth Adaptations for Youth with Autism Spectrum [4]	Emerson, C.,Koob, C.,Sease, K. etal	2025	Qualitative study: stakeholder interviews and adaptation considerations for mHealth tools targeted at youth with ASD	Highlights need for ASD-tailored UX, personalization, and caregiver controls — limited prototyping
5	AI-Enhanced Comprehensive Autism Detection and Support Platform for all ages [5]	Biswas, Koustav and Siddamallappa, Shalini	2025	Preprint describing an AI platform combining detection, chatbot, and modules for daily living	Early/preprint stage — lacks peer review, standardized evaluation

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Gap Identified
6	Human–Robot Companionship: Current Trends and Future Directions [6]	Ahmed, E.,Buruk, O.‘. ,Hamari, J	2024	Literature synthesis on robotic companions and their application, incl. therapeutic contexts	Highlights cost/maintenance and ethics concerns
7	Computer Vision-based Interactive Autism Detection System using Deep Learning [7]	B. Parvej, SMM Alam, F. I. Fahim, MN Pathan, MA Rahaman	2024	Detection of autism traits on image data	Low accuracy, gives incorrect result in few scenario
8	Comparative exploration of machine learning models for enhanced autism detection [8]	C. Nagamani, T. R. Devi, G. G. Babu, V. Sindhu	2024	Compares different ML models on datasets to detect autism	Likely based on questionnaire or structured data

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Gap Identified
9	Towards an automatic early screening system for autism spectrum disorder in toddlers based on eye-tracking [9]	Fernandez-Lanvin, D., Gonzalez-Rodriguez, M., De-Andres, J. et al	2023	This research involves eye-tracking video analysis, feature extraction of gaze patterns, matched-pairs experimental design, ML classification	The drawback is that it needs specialized eye-tracking hardware and has limited robustness across environments
10	Detection of Autism Spectrum Disorder in Children Using Machine Learning [10]	Vakadkar, K., Purkayastha, D. & Krishnan, D.	2021	This paper uses ML classifiers (SVM, Random Forest, Logistic Regression) applied to questionnaire datasets	The drawback is that ML models use structured questionnaire data limiting real world applications

Research Gap(Limitations of existing systems)

- Many individuals with ASD find it difficult to initiate or maintain conversations, making traditional communication-based interventions challenging.
- Sudden changes in daily schedules or unclear routines often lead to stress and anxiety, affecting emotional well-being.
- There is a clear absence of AI-powered systems that combine diagnosis, communication support, routine management, and social skill development into a single platform.

Justification:

These gaps highlight the need for a technology-driven and accessible assistance system that complements therapists rather than replacing them. Our approach provides a comprehensive toolkit for early detection and continuous support through an integrated chatbot, routine tracker, and social skills coach.

Problem Definition

Individuals with Autism Spectrum Disorder (ASD) face significant challenges in early diagnosis, effective communication, and social interaction. Current digital tools are often fragmented, addressing only one aspect of support. This leads to an inconsistent user experience and fragmented care. AutiWise seeks to solve this by developing a single, AI-powered digital toolkit that provides a holistic solution for individuals with ASD and their caregivers.

Proposed Focus Areas -

- Image and Behavioral-based ASD Detection
- Conversational Chatbot for Assistive Communication
- Voice-Enabled Routine Planning
- Social Script Generation for Role-Play and Social Skills Training
- Unified Platform for Long-Term ASD Support

Problem Definition

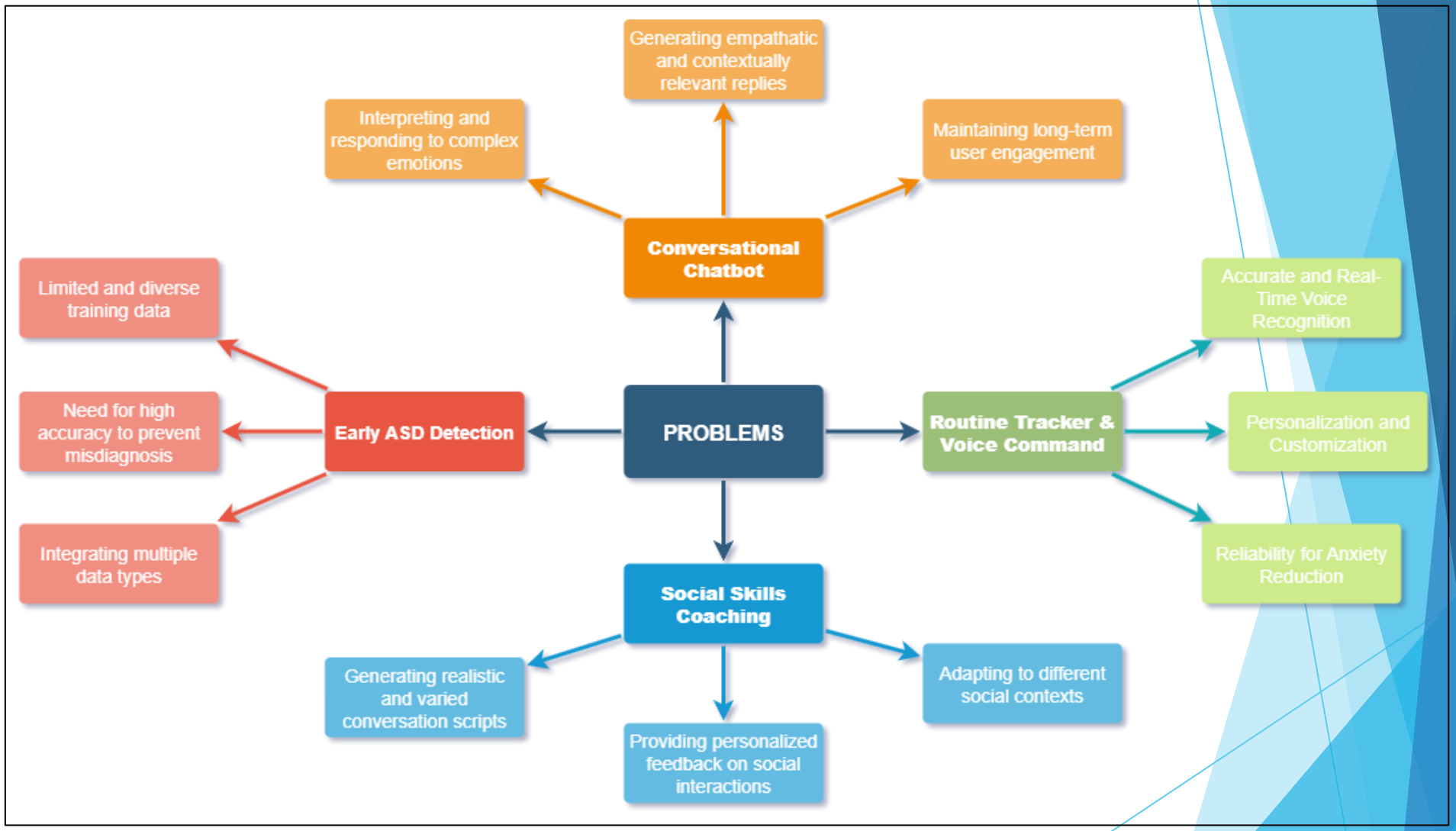


Figure 1: Problem Definition Diagram

Scope

- Design and develop AutiWise, an AI-powered assistance system for individuals with Autism Spectrum Disorder (ASD)
- Support early identification of ASD indicators through behavior and facial analysis.
- Provide an AI conversational chatbot for communication practice and emotional expression.
- Implement a voice-command-based module to create and follow structured daily schedules.
- Develop a social skills coaching module for role-play and real-world conversation practice.
- Ensure user-friendly, accessible, and scalable system design.
- Act as an assistive tool, complementing professional diagnosis and therapy (not a replacement)

Technological Stack

Front – end

- HTML5 / CSS3 / JavaScript: For building responsive and accessible user interfaces

Back – end

- Python
- Flask framework

Models

- LLM model: For chatbot conversations and social script generation
- HaarCascade Classifier (OpenCV): To find precise location of a face in the image
- ResNet50: For ASD detection

Technological Stack

AI-ML/DL

- NLP
- TensorFlow / Keras
- Computer Vision
- CNN

Dataset

- Autistic Chidren Facial Dataset

Trained on ResNet50 model

Source- Kaggle

Proposed system architecture

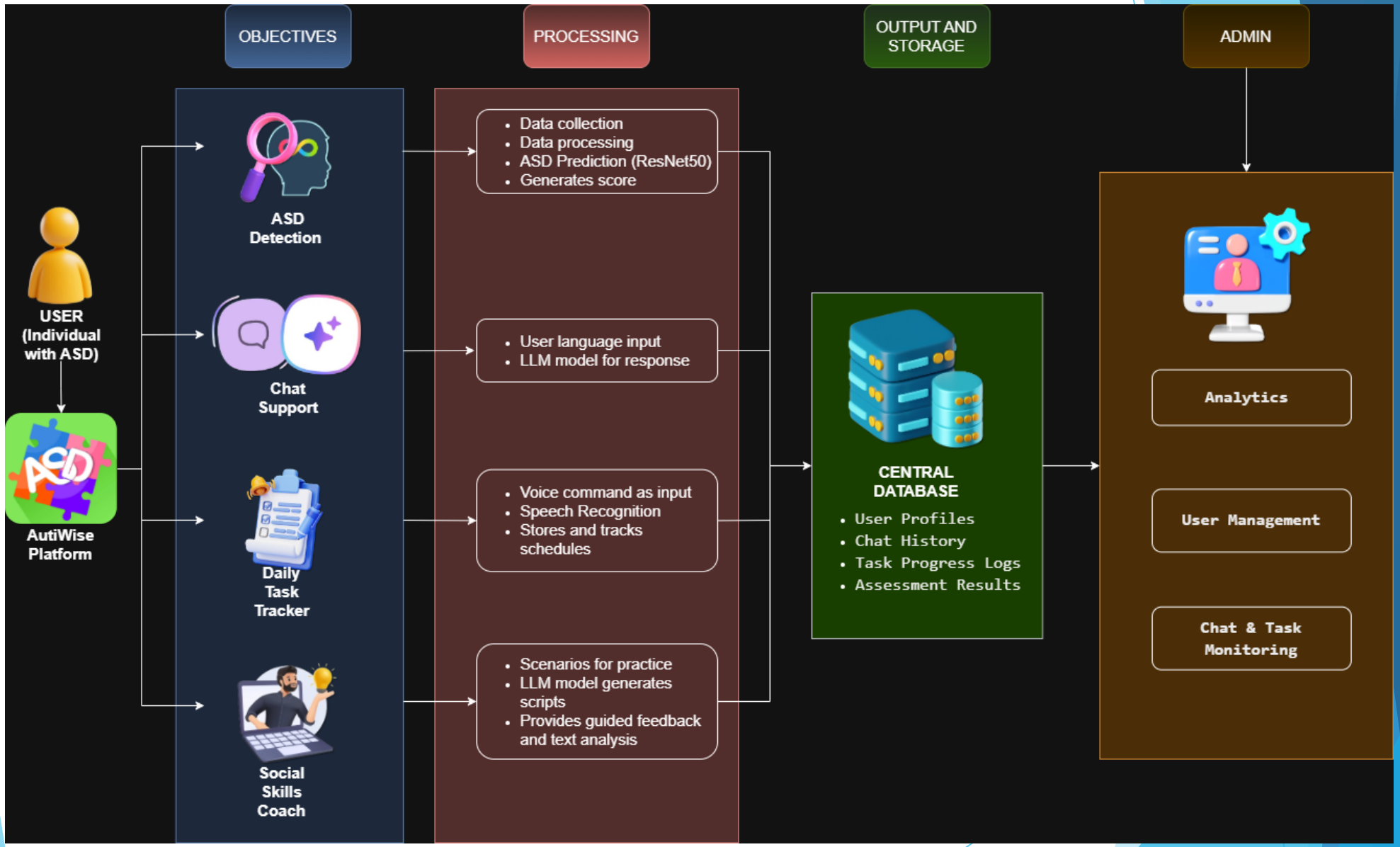


Figure 2: Proposed System Diagram

Prototype Design Demonstration

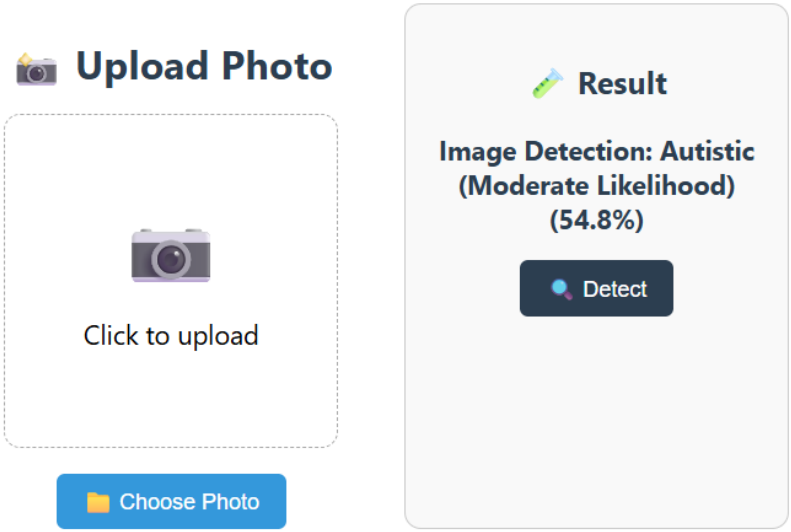


Figure 3: Detection result

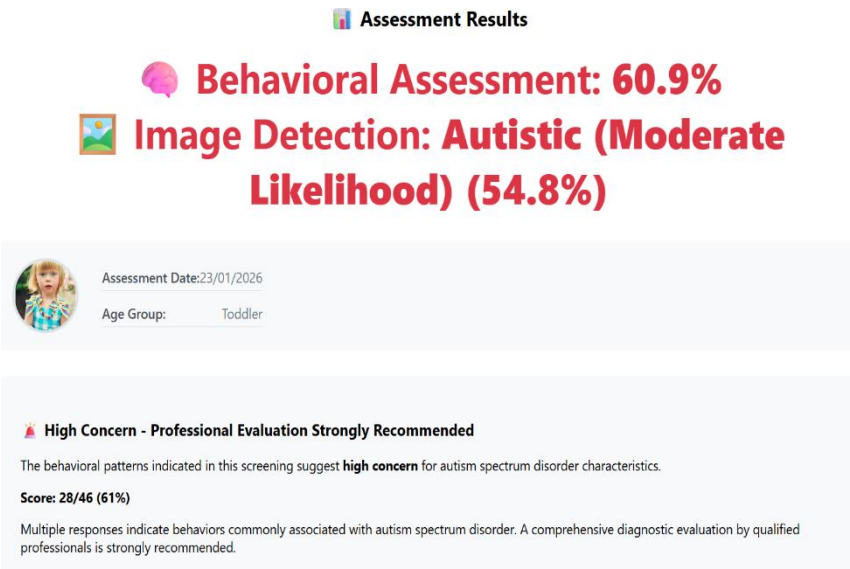


Figure 4: Behavioral result

Prototype Design Demonstration

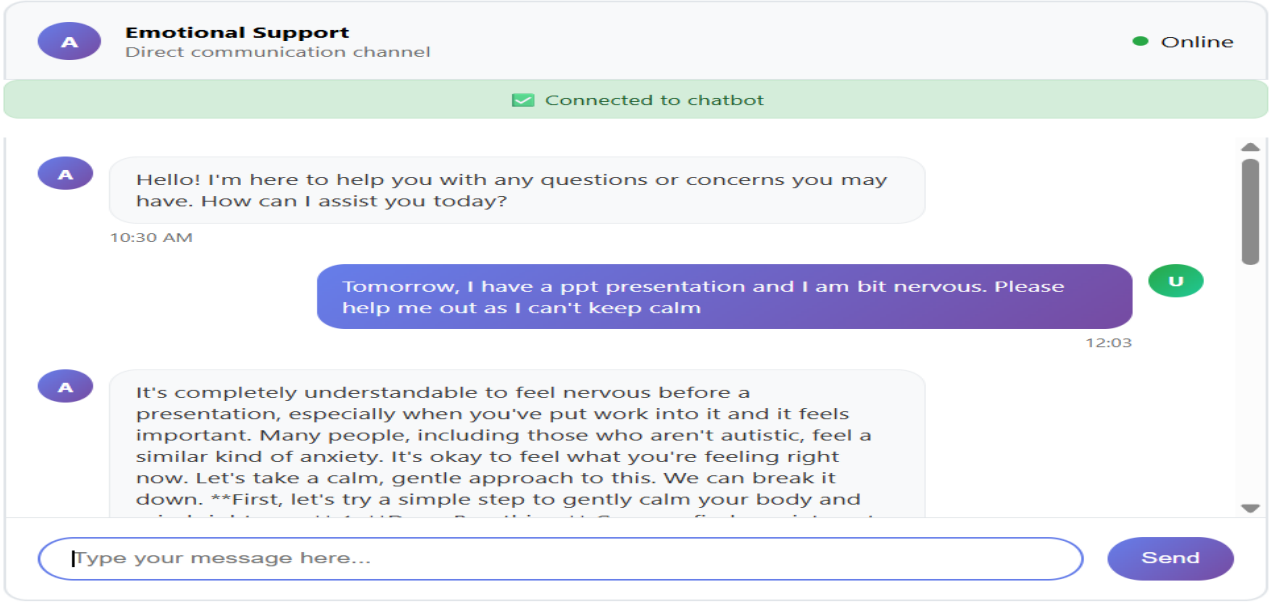


Figure 5: Chatbot result

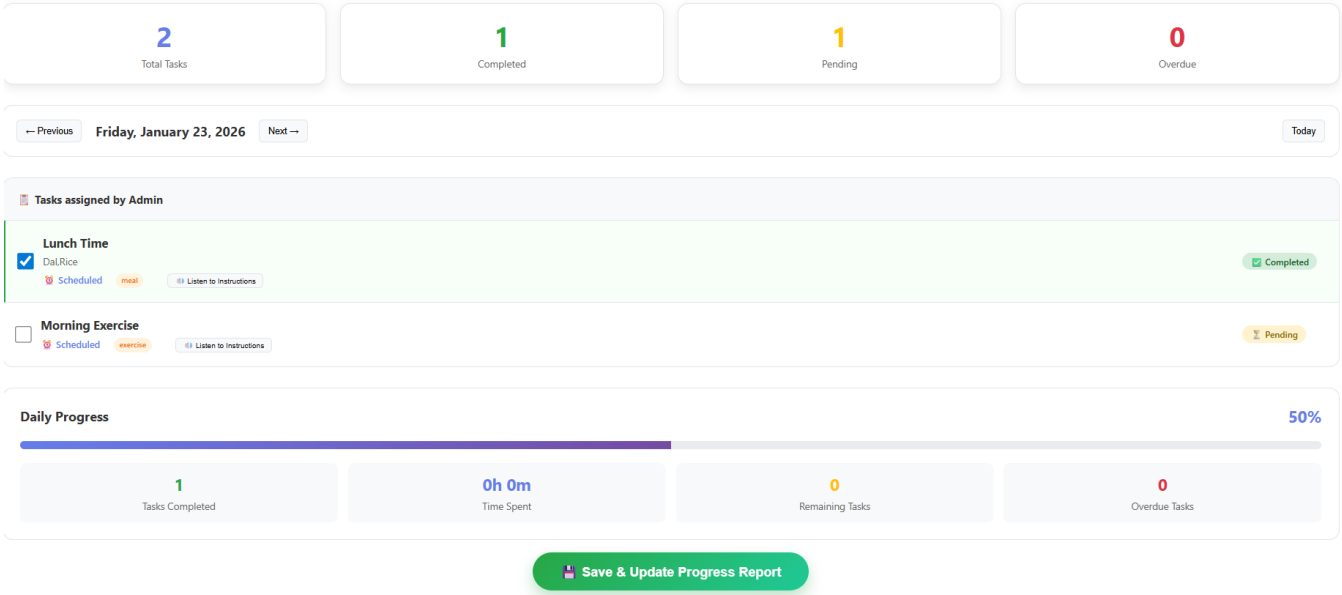
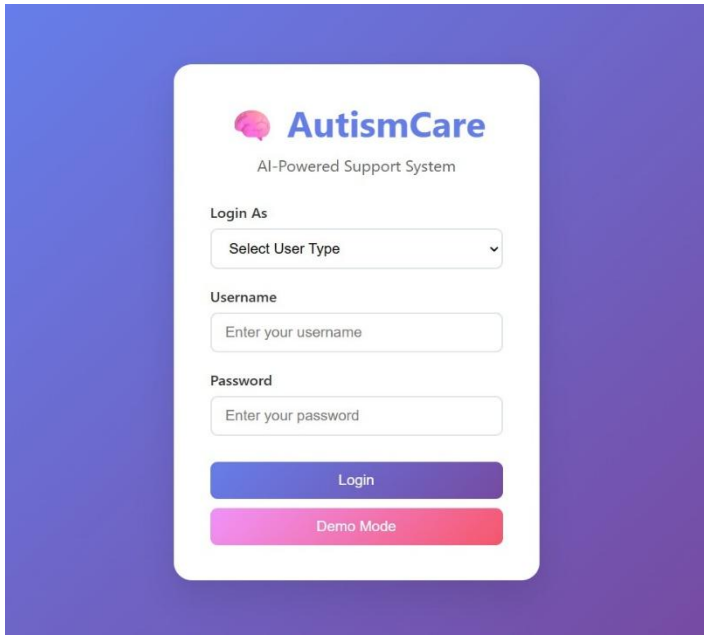
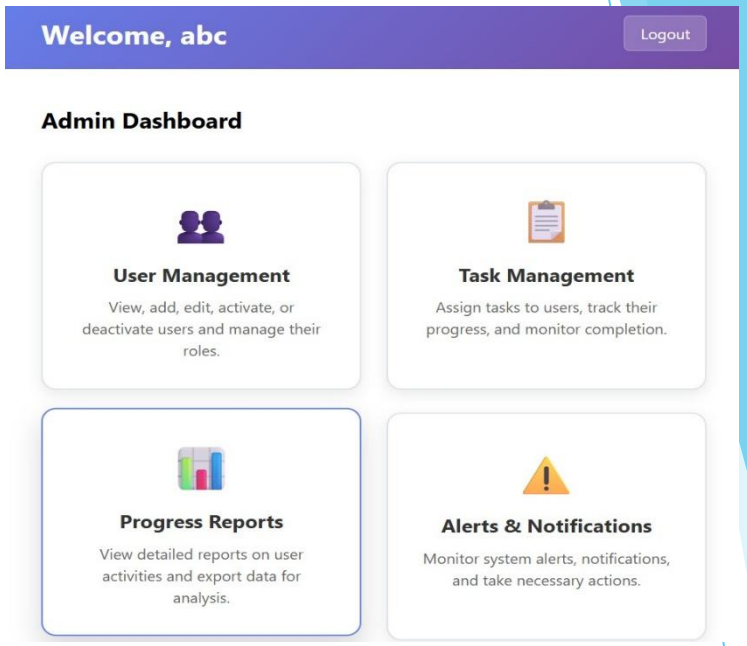
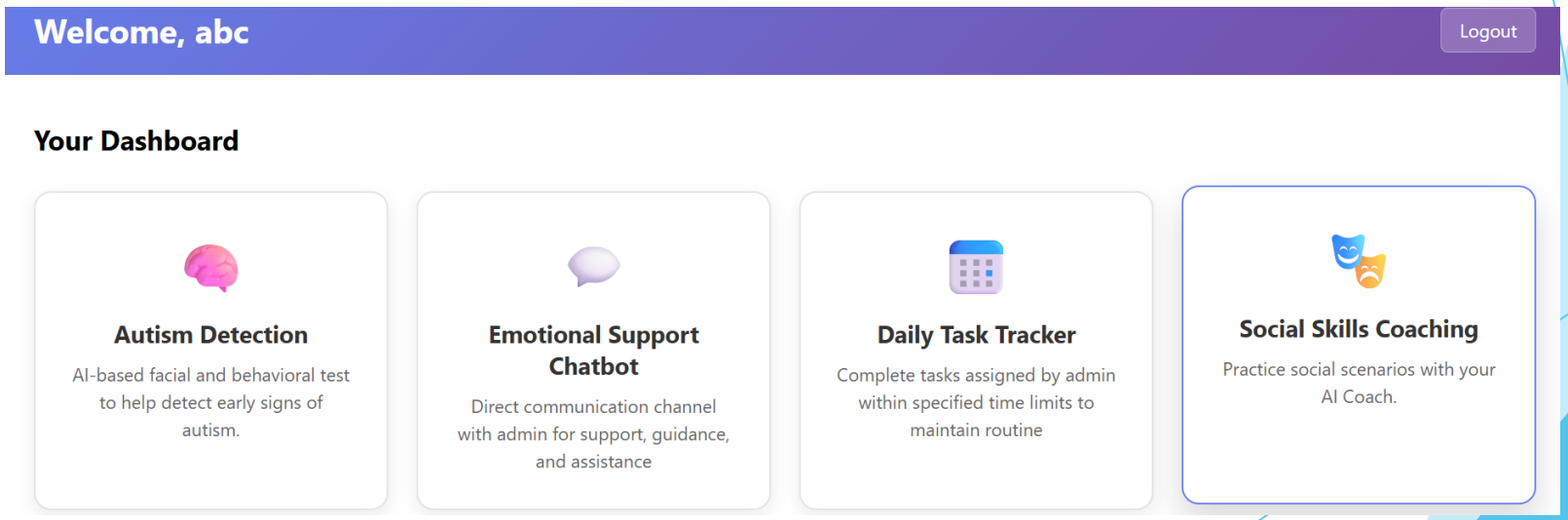


Figure 6: Task tracker result

Prototype Design Demonstration







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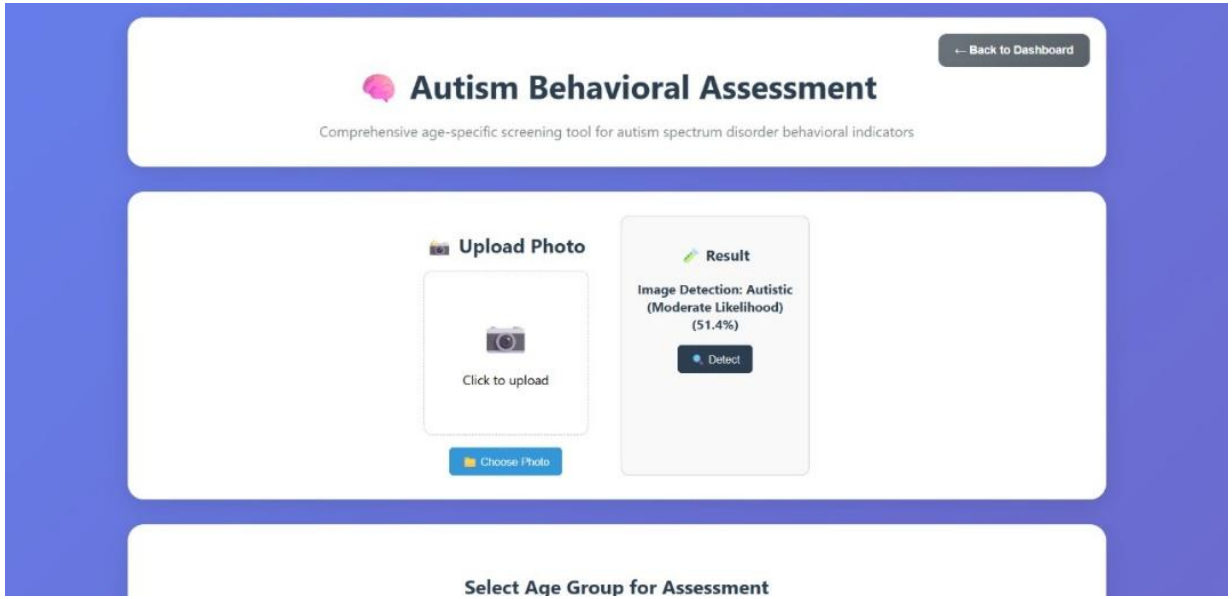

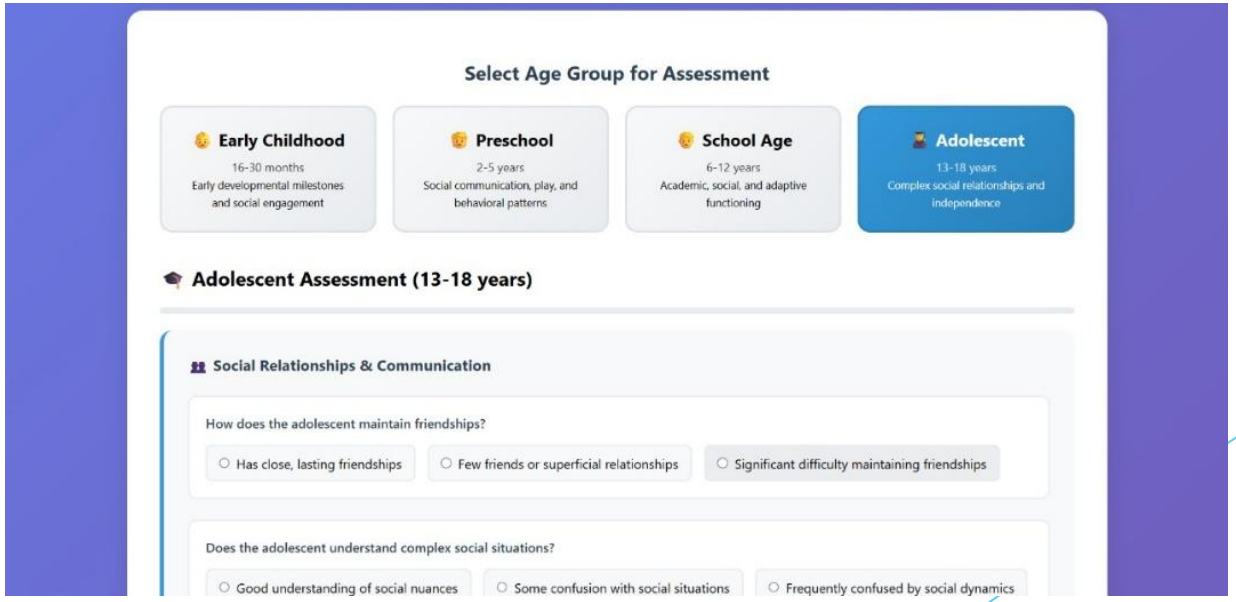

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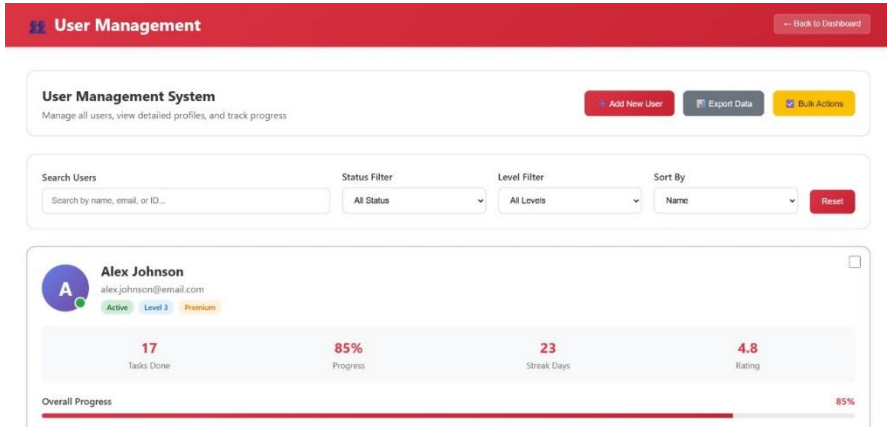

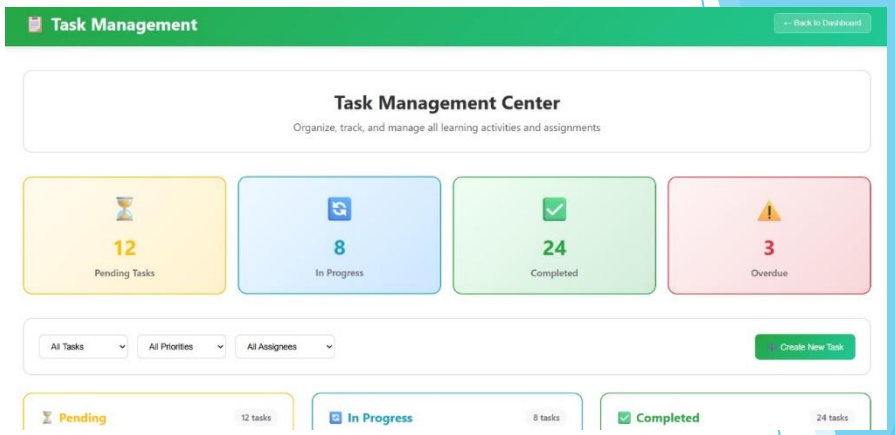

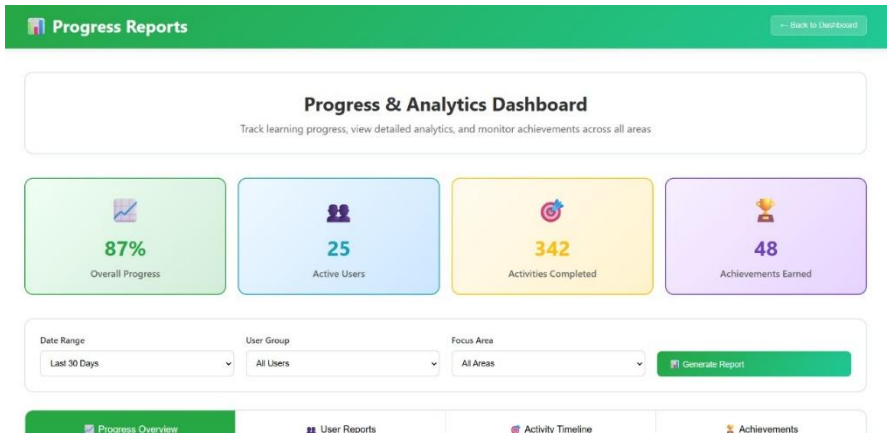

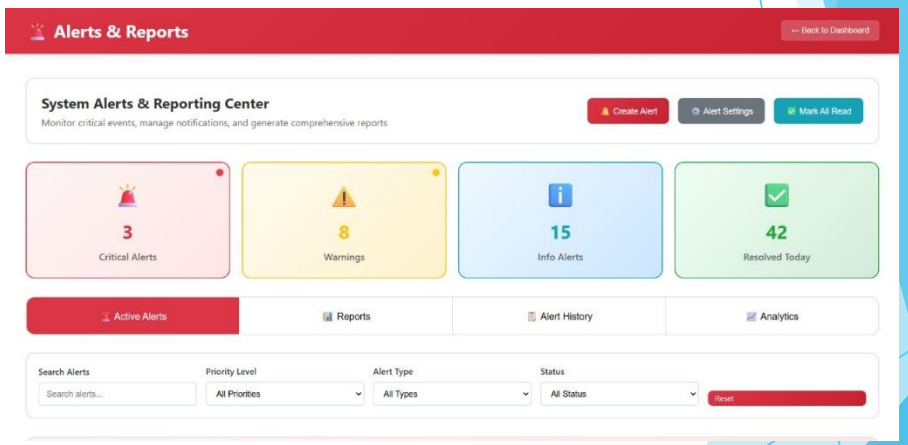

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Thank You...!!